

PHARMA SALES & INVENTORY ANALYTICS

End-to-End Business Intelligence Project

1. Project Overview :

This project demonstrates an end-to-end Business Intelligence solution for a pharmaceutical sales and inventory management system. The objective is to analyze sales performance, inventory levels, product performance, and operational efficiency using Python, SQL, and Power BI.

The project follows a complete analytics workflow, beginning with raw data cleaning, followed by SQL-based business analysis, and finally the creation of an interactive Power BI dashboard for business decision-making.

2. Problem Statement :

Pharmaceutical companies generate large volumes of sales and inventory data every day. Without proper analysis, businesses face challenges such as:

- Inefficient inventory management
- Low-performing products
- Excessive discounts reducing profitability
- Slow delivery performance

- Difficulty identifying top-performing sales representatives
- Lack of centralized business reporting

This project aims to solve these problems through data analytics and visualization.

3. Project Objectives :

1. Clean and prepare raw pharmacy datasets using Python.
2. Perform business analysis using SQL.
3. Build an interactive Power BI dashboard.
4. Analyze sales, profit, inventory, and product performance.
5. Generate actionable business insights for decision-makers.

4. Dataset Description :

The project uses a Star Schema consisting of five datasets.

Sales -

Contains transactional sales records including invoice details, quantity sold, revenue, profit, discounts, payment mode, delivery days, and return status.

Products -

Contains medicine information such as category, manufacturer, dosage form, strength, pack size, MRP, cost price, and shelf life.

Inventory -

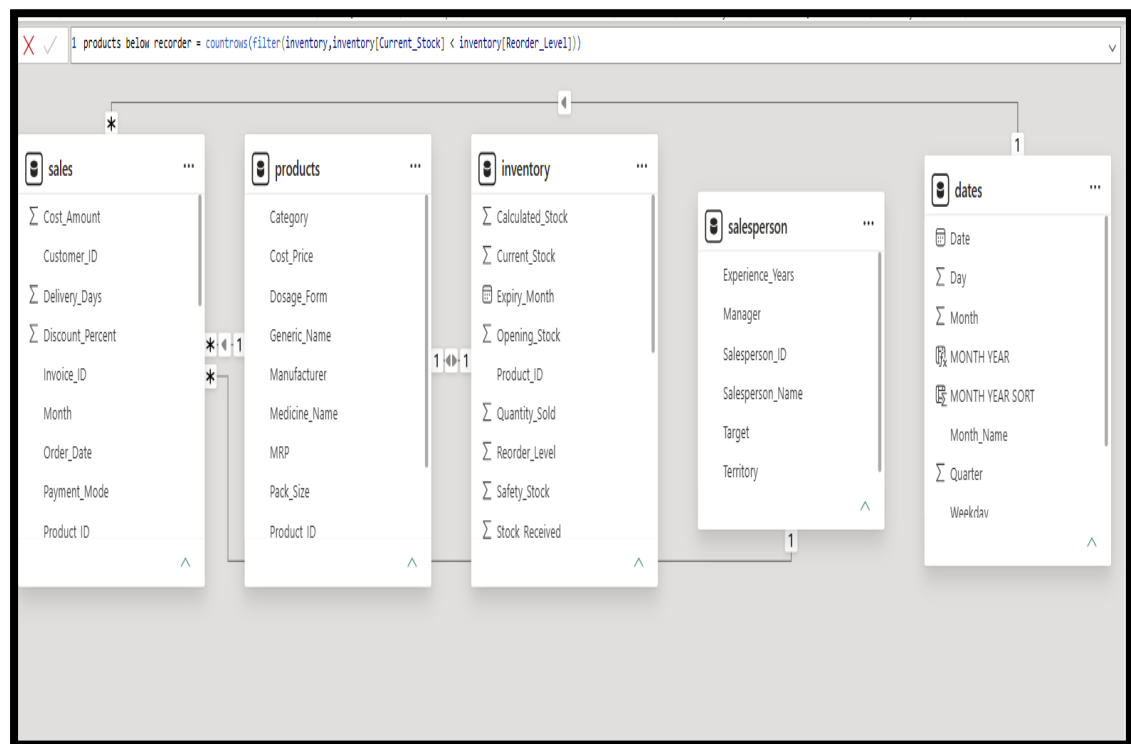
Contains warehouse stock information including opening stock, stock received, quantity sold, current stock, safety stock, reorder level, stock status, and expiry month.

Salesperson -

Contains salesperson details including territory, experience, manager, and sales target.

Date -

Contains calendar information including day, month, quarter, year, weekday, and weekend indicators.



5. Tools & Technologies :

- Python
- Pandas
- SQL Server
- Data querying
- Business analysis
- Power BI
- Data Modeling
- DAX Measures
- Interactive Dashboards
- KPI Cards
- Business Visualizations

6. Python Data Cleaning :

The following preprocessing steps were performed:

- Imported multiple CSV files
- Checked missing values
- Removed duplicate records
- Corrected data types
- Standardized column names
- Converted discount percentage into readable format
- Cleaned inconsistent values
- Exported cleaned datasets for SQL analysis

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jupyter PYTHON NOTEBOOK FOR PHARMA Last Checkpoint: yesterday
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JupyterLab Python 3 (ipykernel)

[6]: Index(['Invoice_ID', 'Order_Date', 'Customer_ID', 'Product_ID',
          'Salesperson_ID', 'Quantity', 'Unit_Price', 'Discount_Percent',
          'Sales_Amount', 'Cost_Amount', 'Profit', 'Payment_Mode',
          'Delivery_Days', 'Return_Flag'],
          dtype='str')

[8]: sales.info

[8]: <bound method DataFrame.info of
Invoice_ID Order_Date Customer_ID Product_ID Salesperson_ID Quantity \
0 INV000001 2025-12-12 CUST050 PROD067 SP12 129
1 INV000002 2025-09-05 CUST035 PROD098 SP20 12
2 INV000003 2024-10-13 CUST047 PROD055 SP16 110
3 INV000004 2025-10-09 CUST003 PROD016 SP13 27
4 INV000005 2024-09-04 CUST108 PROD065 SP20 36
... ..
9995 INV009996 2024-05-17 CUST113 PROD049 SP14 50
9996 INV009997 2024-08-19 CUST114 PROD073 SP01 74
9997 INV009998 2025-01-10 CUST146 PROD007 SP03 75
9998 INV009999 2025-01-27 CUST102 PROD096 SP08 25
9999 INV010000 2024-02-28 CUST196 PROD023 SP09 104

Unit_Price Discount_Percent Sales_Amount Cost_Amount Profit \
0 182.45 0.00 23536.05 14227.41 9308.64
1 113.41 0.00 1360.92 1123.92 237.00
2 39.60 0.00 4356.00 2888.60 1467.40
3 47.33 0.00 1277.91 1004.67 273.24
4 98.37 0.00 3541.32 2146.68 1394.64
... ..
9995 215.14 0.00 10757.00 7431.00 3326.00
9996 220.00 0.05 15466.00 10885.40 4580.60
9997 210.04 0.00 15753.00 10106.25 5646.75
9998 62.46 0.00 1561.50 956.00 605.50
9999 67.67 0.00 7037.68 4409.60 2628.08

Payment_Mode Delivery_Days Return_Flag
0 Credit Card 2 1
1 Credit Card 5 0
2 Bank Transfer 4 0
3 Credit Card 5 0
4 Cheque 6 0
... ..
9995 Credit Card 2 0
9996 Credit Card 1 0
9997 Credit Card 2 0
9998 Bank Transfer 3 0
9999 Cheque 3 0

[10000 rows x 14 columns]>

[6]: sales['Discount_Percent'] = sales['Discount_Percent'] * 100
```

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JupyterLab Python 3 (ipykernel)

[1]: import pandas as pd
import numpy as np

[3]: sales = pd.read_csv("fact_sales.csv")
sales.head(10)

[3]: Invoice_ID Order_Date Customer_ID Product_ID Salesperson_ID Quantity Unit_Price Discount_Percent Sales_Amount Cost_Amount Profit Payment_Mode C
0 INV000001 2025-12-12 CUST050 PROD067 SP12 129 182.45 0.00 23536.05 14227.41 9308.64 Credit Card
1 INV000002 2025-09-05 CUST035 PROD098 SP20 12 113.41 0.00 1360.92 1123.92 237.00 Credit Card
2 INV000003 2024-10-13 CUST047 PROD055 SP16 110 39.60 0.00 4356.00 2888.60 1467.40 Bank Transfer
3 INV000004 2025-10-09 CUST003 PROD016 SP13 27 47.33 0.00 1277.91 1004.67 273.24 Credit Card
4 INV000005 2024-09-04 CUST108 PROD065 SP20 36 98.37 0.00 3541.32 2146.68 1394.64 Cheque
5 INV000006 2025-05-23 CUST142 PROD036 SP01 118 217.36 0.00 25648.48 14944.70 10703.78 Credit Card
6 INV000007 2025-03-24 CUST075 PROD067 SP20 59 182.45 0.00 10764.55 6507.11 4257.44 Cheque
7 INV000008 2024-08-31 CUST012 PROD049 SP10 86 215.14 0.05 17576.94 12781.32 4795.62 Bank Transfer
8 INV000009 2024-05-13 CUST067 PROD080 SP07 18 54.57 0.00 982.26 698.76 283.50 Bank Transfer
9 INV000010 2025-04-10 CUST020 PROD082 SP10 29 60.60 0.00 1757.40 1172.47 584.93 Bank Transfer

[4]: sales.tail()

[4]: Invoice_ID Order_Date Customer_ID Product_ID Salesperson_ID Quantity Unit_Price Discount_Percent Sales_Amount Cost_Amount Profit Payment_Mode
9995 INV009996 2024-05-17 CUST113 PROD049 SP14 50 215.14 0.00 10757.00 7431.00 3326.00 Credit Card
9996 INV009997 2024-08-19 CUST114 PROD073 SP01 74 220.00 0.05 15466.00 10885.40 4580.60 Credit Card
9997 INV009998 2025-01-10 CUST146 PROD007 SP03 75 210.04 0.00 15753.00 10106.25 5646.75 Credit Card
9998 INV009999 2025-01-27 CUST102 PROD096 SP08 25 62.46 0.00 1561.50 956.00 605.50 Bank Transfer
9999 INV010000 2024-02-28 CUST196 PROD023 SP09 104 67.67 0.00 7037.68 4409.60 2628.08 Cheque

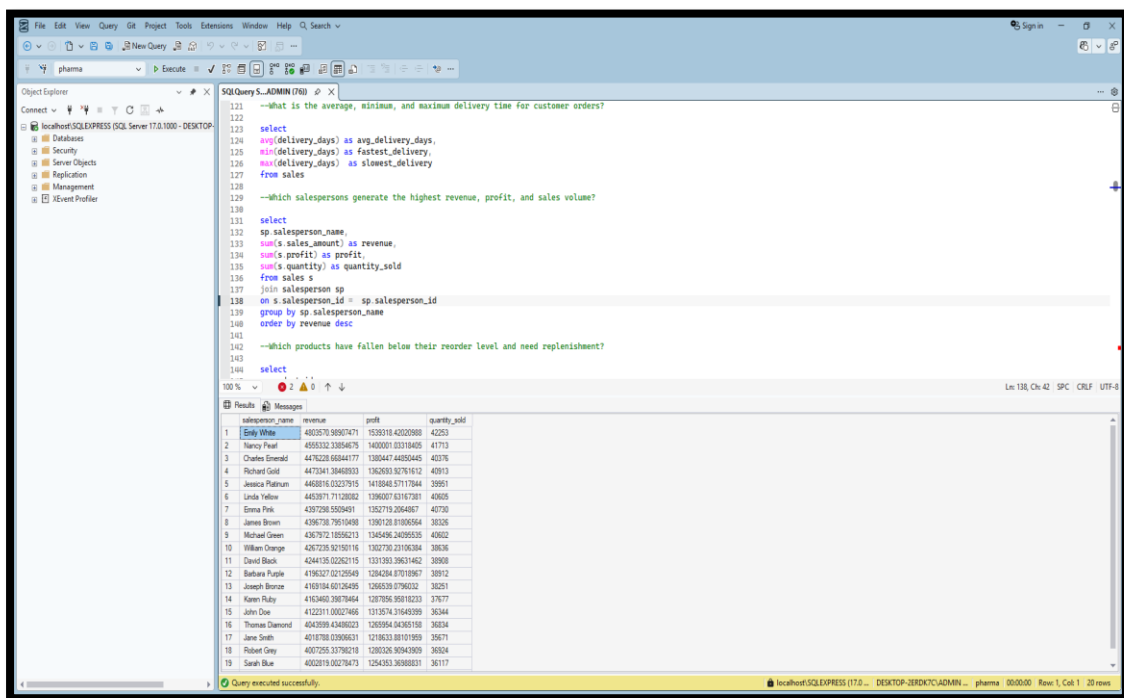
[5]: sales.shape

[5]: (10000, 14)
```

7. SQL Analysis :

SQL queries were written to answer key business questions and support data-driven decision-making.

- Total Revenue
- Total Profit
- Monthly Sales Trend
- Category-wise Sales
- Manufacturer Performance
- Product Profitability
- Payment Mode Analysis
- Return Analysis
- Salesperson Performance
- Inventory Analysis
- Warehouse Stock Analysis
- Low Stock Products
- Product-wise Sales
- Quarterly Performance



The screenshot displays the SQL Server Enterprise Manager interface. The 'Object Explorer' on the left shows the 'pharma' database. The 'Query Window' on the right contains three SQL queries. The first query calculates average, minimum, and maximum delivery times. The second query lists salespersons by revenue, profit, and sales volume. The third query identifies products that need replenishment. The 'Results' pane at the bottom shows the output of the second query, listing salespersons and their performance metrics.

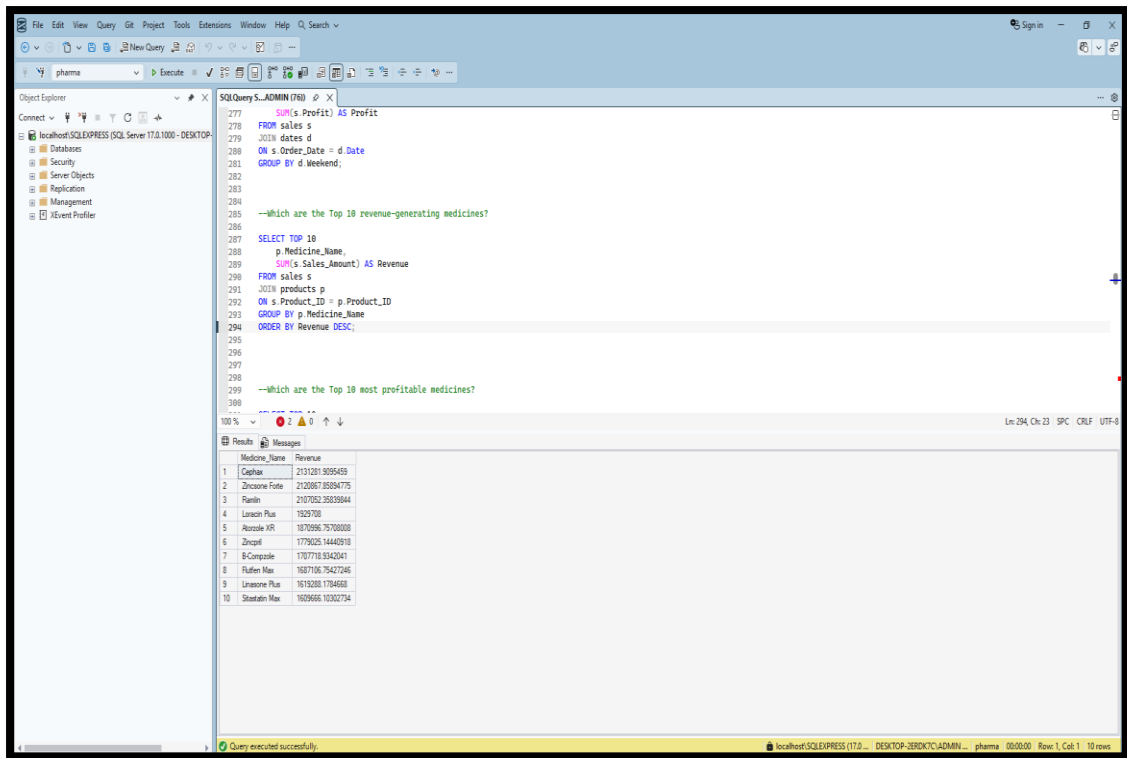
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--What is the average, minimum, and maximum delivery time for customer orders?
select
avg(delivery_days) as avg_delivery_days,
min(delivery_days) as fastest_delivery,
max(delivery_days) as slowest_delivery
from sales

--Which salespersons generate the highest revenue, profit, and sales volume?
select
sp.salesperson_name,
sum(s.sales_amount) as revenue,
sum(s.profit) as profit,
sum(s.quantity) as quantity_sold
from sales s
join salesperson sp
on s.salesperson_id = sp.salesperson_id
group by sp.salesperson_name
order by revenue desc

--Which products have fallen below their reorder level and need replenishment?
select
```

	salesperson_name	revenue	profit	quantity_sold
1	Emily White	4003570.98907471	1539318.42020988	42253
2	Nancy Pearl	4595332.33954579	1400001.03119409	41713
3	Charles Emerald	4476228.65844177	1380447.44955405	40276
4	Richard Gold	4473341.38488933	1362693.92761612	40913
5	Jessica Platinum	4408816.0327915	1418848.57117844	39951
6	Linda Yellow	4403871.71120582	1396007.63167381	40005
7	Emma Pink	4397295.55059431	1352719.2054367	40700
8	James Brown	4394735.76515498	1390128.81085564	38226
9	Michael Green	4367972.18556213	1345495.24055305	40602
10	William Orange	4267235.92190116	1302730.23106384	38036
11	David Black	4244135.02262115	1331393.39631462	38908
12	Barbara Purple	4196327.02125549	1284294.87018967	38912
13	Joseph Brown	4189184.60125495	1265339.8796032	38251
14	Karen Ruby	4163460.39870464	1287856.95918223	37677
15	John Doe	4122311.00027466	1313574.31649399	36344
16	Thomas Diamond	4043939.43480023	1265954.04363158	36034
17	Jane Smith	4018788.03906531	1218633.8017059	35671
18	Robert Grey	4007555.37378118	1280326.30430309	36024
19	Sarah Blue	4002019.00278473	1254363.36388031	36117

Query executed successfully

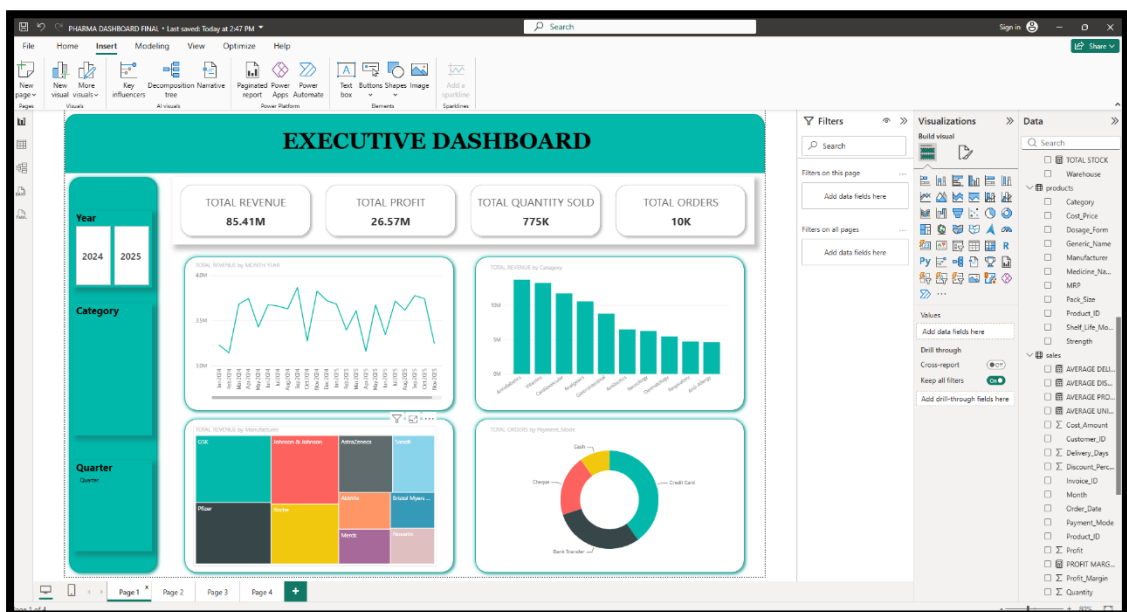


8. Power BI Dashboard

The dashboard consists of four interactive pages.

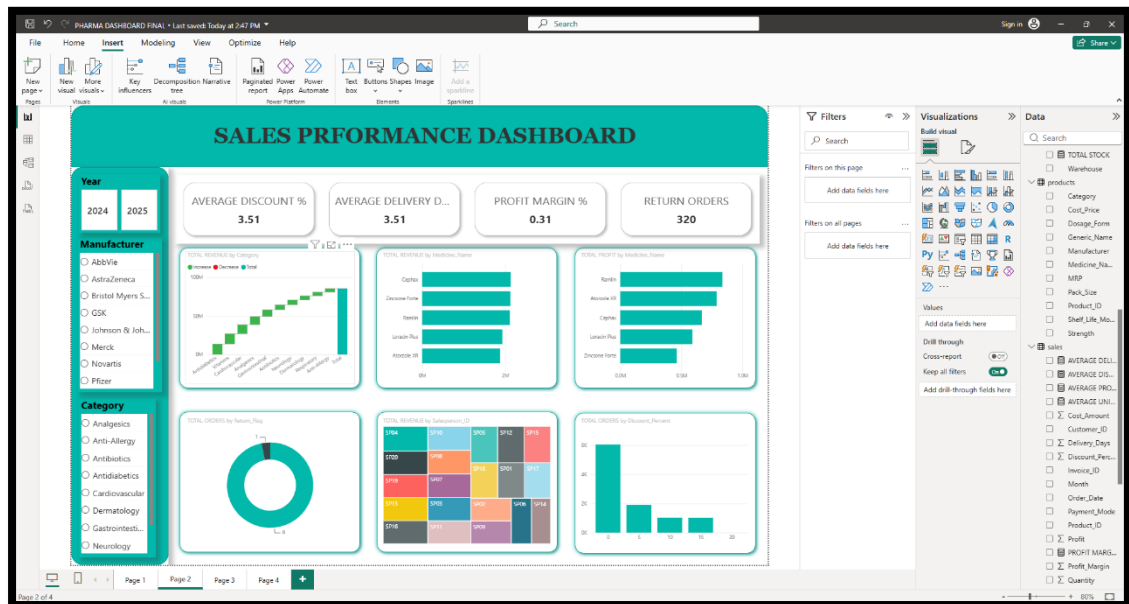
Page 1 – Executive Dashboard

Provides an overall business summary using KPIs and revenue trends.



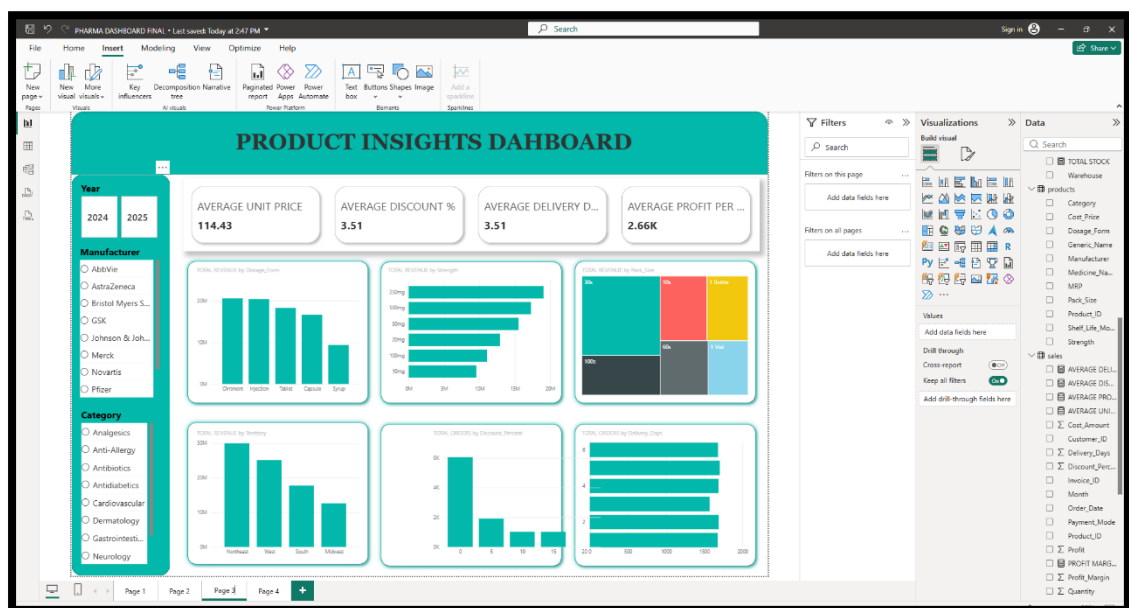
Page 2 – Sales Performance

Analyzes sales performance by products, manufacturers, payment mode, discounts, and sales representatives.



Page 3 – Product Insights

Focuses on product categories, dosage forms, strengths, manufacturers, pricing, and product performance.



Provides inventory monitoring including warehouse stock, stock status, reorder analysis, safety stock, and current inventory levels.



1. Identify top-selling medicines.
2. Monitor profit-generating products.
3. Evaluate manufacturer performance.
4. Analyze discount utilization.
5. Compare payment methods.
6. Track inventory availability.
7. Identify products requiring replenishment.
8. Monitor warehouse-wise stock distribution.
9. Evaluate salesperson performance.
10. Support data-driven business decisions.

10.Business Recommendations :

1. Increase inventory for consistently high-demand medicines.
2. Review products with low profitability.
3. Optimize discount strategies to improve margins.
4. Monitor inventory regularly to prevent stock shortages.
5. Improve delivery performance through logistics optimization.
6. Focus marketing efforts on high-performing product categories.

11.Skills Demonstrated :

- Data Cleaning
- Data Transformation
- SQL Query Writing
- Data Modeling
- Star Schema Design
- DAX Measures
- Business Intelligence
- Dashboard Development
- Data Visualization
- Business Analysis
- Problem Solving